



Please note that the links in the PEARL logotype above are “live” and can be used to direct your web browser to our site or to open an e-mail message window addressed to ourselves.

To view our item listings on eBay, [click here](#).

To see the feedback we have left for our customers, [click here](#).

This document has been prepared as a public service . Any and all trademarks and logotypes used herein are the property of their owners.

It is our intent to provide this document in accordance with the stipulations with respect to “fair use” as delineated in Copyrights - Chapter 1: Subject Matter and Scope of Copyright; Sec. 107. Limitations on exclusive rights: Fair Use.

Public access to copy of this document is provided on the website of Cornell Law School at <http://www4.law.cornell.edu/uscode/17/107.html> and is here reproduced below:

Sec. 107. - Limitations on exclusive rights: Fair Use

Notwithstanding the provisions of sections 106 and 106A, the fair use of a copyrighted work, including such use by reproduction in copies or phono records or by any other means specified by that section, for purposes such as criticism, comment, news reporting, teaching (including multiple copies for classroom use), scholarship, or research, is not an infringement of copyright. In determining whether the use made of a work in any particular case is a fair use the factors to be considered shall include:

- 1 - the purpose and character of the use, including whether such use is of a commercial nature or is for nonprofit educational purposes;
- 2 - the nature of the copyrighted work;
- 3 - the amount and substantiality of the portion used in relation to the copyrighted work as a whole; and
- 4 - the effect of the use upon the potential market for or value of the copyrighted work.

The fact that a work is unpublished shall not itself bar a finding of fair use if such finding is made upon consideration of all the above factors

A Model of Loudspeaker Driver Impedance Incorporating Eddy Currents in the Pole Structure*

JOHN VANDERKOOY**

Audio Research Group, Department of Physics, University of Waterloo, Waterloo, Ontario N2L 3G1, Canada

Measurements of electrodynamic drivers show a blocked-cone impedance which is often dominated by a $\sqrt{j\omega}$ component. A simple theory shows that eddy currents in the iron pole structure are responsible for this behavior. A more complex theory shows a transition between such semi-inductive behavior at high frequencies to a normal inductive behavior at low frequencies, but a vast reduction of the effective permeability of the iron must be invoked to explain the measured data. This reduction is due to the effective demagnetizing gaps in the pole structure. Other aspects of the eddy currents concern the asymmetric drag force on a slotted aluminum voice-coil former, the direct thermal heating of the pole structure by the eddy currents, the electrical compensation of the $\sqrt{j\omega}$ semi-inductive impedance, and the current waveforms expected when a step or impulse voltage is applied to the driver.

0 INTRODUCTION

In typical simple treatments of electrodynamic loudspeakers, equivalent circuits used to model the electric impedance consider the voice coil as having a resistance and an inductance, while the coil gives a back EMF proportional to cone velocity. The acoustomechanical aspects of the cone motion will be reflected in the impedance, and the loudspeaker output can be predicted from the electric input in the range of frequencies for which the cone moves as a rigid assembly.

The acoustic, mechanical, and electrical aspects can be lumped together into a motional electric impedance $B^2 l^2 Z_M$ [1], in which B is the field in the coil gap, l is the total voice-coil wire length, and Z_M has units of meters per newton-second (mechanical mobility ohms), which relates the coil velocity to the force on it. If the voice coil is blocked, the motional electric impedance becomes zero, and the electric impedance should be just a series resistor and inductor. Measurements on actual drivers presented later do not support such a simple view, and this fact has been noted by many workers over the years. It is generally recognized that eddy currents flowing in the solid iron pole structure

will modify the electric impedance [2], [3]. This is commonly dealt with by employing lossy inductors in the model, that is, inductors that may have a series resistance but also have a shunting parallel resistance. By employing enough such dissipative inductors the model can be made to fit the impedance data as well as necessary over a restricted but often adequate frequency range, at the expense of an increased number of parameters.

This paper seeks to work out the effects of the eddy currents in more precise fashion, and while the theory is more complex than simple lossy inductors, the final outcome is increased understanding of the effect of the eddy currents, a reduced number of required parameters to describe the impedance accurately, and new insights into the power-handling ability of electrodynamic drivers. Although the role of eddy currents may have been analyzed many times over the years, the author is aware of only one study [4], which emerges with the $\sqrt{jf}$ impedance behavior that will be shown to be a dominant characteristic of eddy currents in the iron pole structure.

1 SOME EXAMPLES AND A SIMPLE THEORY

A typical measured impedance for an unmounted 8-in (203-mm) low-frequency driver unit is shown in Fig. 1. The impedance is measured by a maximum-length sequence (MLS) technique, as outlined in [5]. The input MLS excitation signal is fed to the driver

* Presented at the 84th Convention of the Audio Engineering Society, Paris, France, 1988 March 1–4; revised 1989 January 5.

** Member of (GWP)², the Guelph-Waterloo Program for Graduate Work in Physics.

through a 30- Ω resistor, and the voltage across the driver, relative to the input signal, is determined as a function of frequency by a computer using a cross-correlation algorithm [6] and the discrete Fourier transform. A correction is applied for the series resistor, and the resulting magnitude and phase of the impedance are plotted as shown. The measurements are done with proper filters to prevent aliasing, and the phase response of these filters is removed by reference normalization, resulting in accurate impedance functions. It is easy to manipulate the computer data for further processing.

While the woofer unit whose impedance is shown in Fig. 1 would probably not be used above 1 kHz, its impedance changes markedly even over this range. Note that the phase is close to $\pi/4$, or 45° , over much of the plot, while the magnitude appears to be proportional to the square root of frequency. At very low frequencies the free-air resonance of the driver manifests itself, as shown in Fig. 2. The dc resistance of the voice coil is 5.7 Ω , and since this resistance is always in series with the total impedance (except perhaps at very high frequencies due to distributed capacitance), the computer can be programmed to remove it from the plotted impedance desired, to obtain just the portion of impedance related to Faraday-induced electromotive force (EMF). This EMF has two components, one created by the motion of the voice coil, the other from the flux change due to eddy currents in the iron central pole.

In an effort to measure just the eddy-current effects, the driver cone can be blocked while the impedance is measured. Fig. 3 shows the impedance over the same frequency range as Fig. 2, but with the cone blocked mechanically. Most of the resonant effects are gone, but small resonances often remain at higher frequencies due to the greatly increased restoring force. It is not the aim of this paper to focus on the driver resonance, which has been modeled by many authors, but rather on the underlying impedance with the cone held motionless. The blocked-cone impedance is significant, for if it is not resistive, there will be an influence on the measurement of the driver parameters when using a test box. This topic will be touched on later.

An even better way to measure a blocked-cone

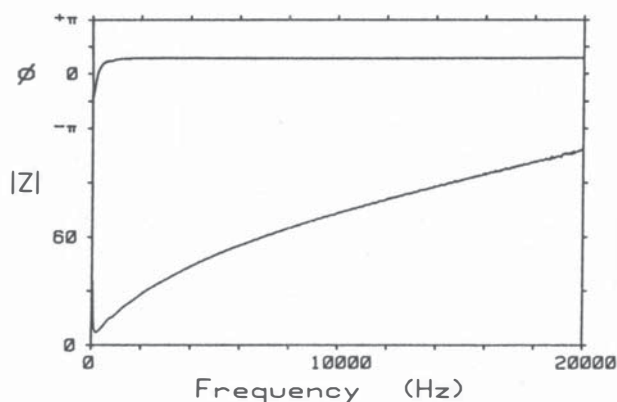


Fig. 1. Impedance magnitude and phase angle for unmounted 8-in (203-mm) woofer driver, measured using MLS techniques. The dominant $\sqrt{f}$ magnitude and 45° phase angle are the main subjects of this paper.

impedance is to deenergize the driver unit's magnet, or to use just the pole structure, unenergized magnet, and voice-coil assembly of an unassembled driver. Fig. 4 shows the impedance of the latter structure for a 1-in (25.4-mm)-diameter central pole, with the dc re-

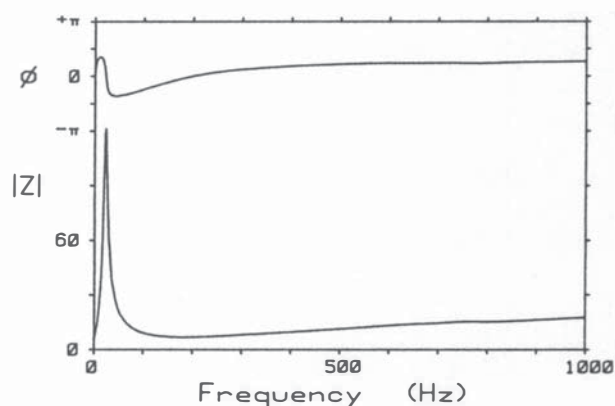


Fig. 2. Impedance magnitude and phase angle for woofer unit of Fig. 1, but measured on a lower frequency range. The fundamental resonance is now quite visible.

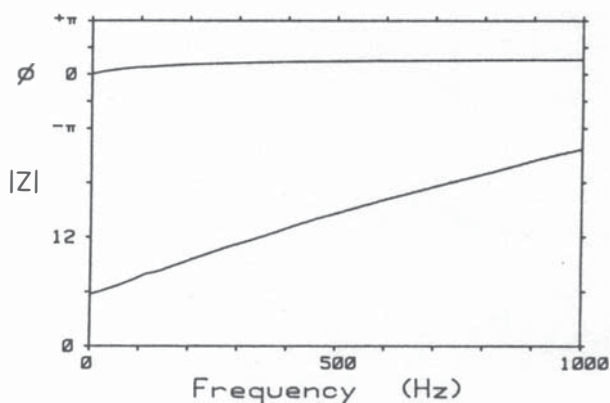


Fig. 3. Impedance magnitude and phase angle for woofer unit of Figs. 1 and 2, but with cone mechanically blocked to prevent motion. Some resonant effects remain because the blocking of the cone motion is not complete.

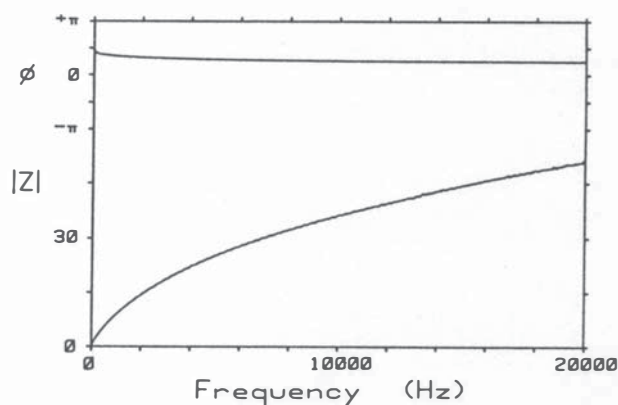


Fig. 4. Impedance magnitude and phase angle for driver unit motor structure having a 1-in (25.4-mm)-diameter voice coil and unenergized magnet. There is no voice-coil motion, but the environment of the voice coil is identical to that of a normal driver. The dc coil resistance of 4.6 Ω has been subtracted from the actual impedance, thus the displayed curve shows only the effect of the induced voltage in the voice coil. Note the pronounced $\sqrt{f}$ behavior of the magnitude.

sistance of 4.6Ω removed by the computer. Note again the almost perfect square root behavior and 45° phase angle, except at very low frequencies, where the impedance is linear with frequency and the phase angle approaches 90° . This inductive portion would not be visible had the dc resistance not been subtracted from the total impedance.

In attempting a simplified theory of the blocked-cone impedance, we note that the central pole of conducting iron will have eddy currents induced in it from the excitation currents in the voice coil. A well-known elementary analysis of such eddy currents (given later) shows that they are confined roughly within a skin depth δ from the surface, given by

$$\delta = \sqrt{\frac{2}{\mu\sigma\omega}} \quad (1)$$

where μ is the magnetic permeability of the pole material, σ is its conductivity, and ω is the angular frequency of the excitation currents. For a relative permeability of 3000 ($\mu = 3000\mu_0 = 3 \times 10^3 \times 4\pi \times 10^{-7}$ H/m) and a conductivity σ of $10^7 \Omega^{-1}/\text{m}$, the skin depth δ , even at 20 Hz, is only 0.65 mm. This is much smaller than the radius of the pole, which is typically near 10 mm. Hence we shall simplify the initial analysis by considering how an oscillating magnetic field, tangential to a semi-infinite conducting medium, propagates into the medium. The cylindrical curvature of the voice coil and pole structure will be dealt with later.

The geometry of the model is shown in Fig. 5, with the yz plane representing the surface of the conducting medium, which extends into the half-space specified by increasing x . We regard the voice coil as an oscillating current sheet with the wires oriented in the z direction, which produces an oscillating magnetic field H at the surface in the y direction. Although there is an air gap between the coil and the pole, it is small and will initially be neglected, in spite of its being responsible for a normal inductive component in series with the coil impedance.

Maxwell's equations govern the propagation of electromagnetic waves, and in terms of the electric field E , displacement D , field induction B , magnetic intensity H , and current density J , we have

$$\nabla \times E = - \frac{\partial B}{\partial t} \quad (2)$$

$$\nabla \times H = J + \frac{\partial D}{\partial t} \quad (3)$$

and since there is no free electric charge inside a conductor, and magnetic monopoles do not exist, in addition the two field divergences are zero:

$$\nabla \cdot E = 0 \quad (4)$$

$$\nabla \cdot B = 0 \quad (5)$$

The current density J in the conducting pole material is related to the electric field by

$$J = \sigma E \quad (6)$$

where σ is the scalar conductivity. In reality the static magnetic field in a driver pole structure causes Hall effects in the current flow so that Eq. (6) is not exactly correct, but we ignore such effects here. The author is familiar with an earlier analysis of eddy currents in metals at low temperatures [7] in which the Hall effect and the eddy currents interplay to explain the measured behavior.

In addition to the preceding equations, we need the relationship

$$B = \mu H \quad (7)$$

in which μ represents the magnetic permeability, which is often several thousand times larger in iron than in free space, for which $\mu_0 = 4\pi \times 10^{-7}$ H/m. Also, the displacement current density term $\partial D/\partial t$ in Eq. (3) is usually ignored in conducting media, since this term is less than J by a factor $\epsilon\omega/\sigma$, where ϵ is the electric permittivity of the material and ω is the angular frequency. Typically $\epsilon\omega/\sigma < 10^{-13}$, even at the highest audio frequencies. The loss of the displacement current term is very significant, for it means that normal electromagnetic waves cannot propagate in strongly conducting media.

Using Eqs. (2)–(7) and the vector identity (for H or E)

$$\nabla \times (\nabla \times H) = \nabla(\nabla \cdot H) - \nabla^2 H \quad (8)$$

where ∇^2 is the Laplacian operator, it is easily shown that

$$\nabla^2 H = \mu\sigma \frac{\partial H}{\partial t} \quad (9)$$

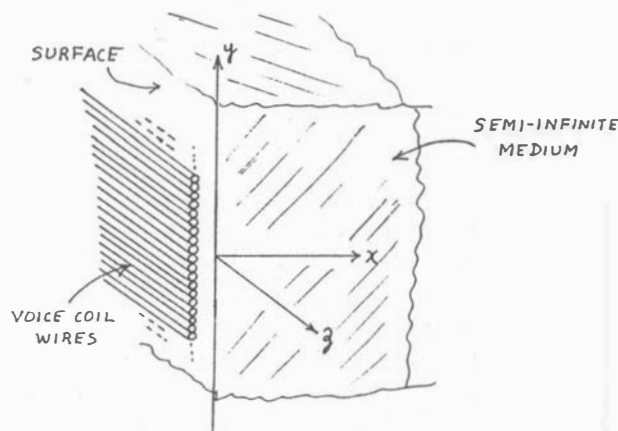


Fig. 5. Geometry of semi-infinite model. The magnetic medium has its surface in the yz plane and extends infinitely in the $+x$ direction. The voice coil represents a current sheet carrying current in the z direction, which creates a magnetic field at the surface of the medium in the y direction.

and an identical equation results for E .

Eq. (9) has the form of a diffusion equation, which has only a single time derivative, whereas a normal wave equation has a double time derivative, while the double spatial derivatives of the Laplacian ∇^2 are the same for both types of equations.

The geometry of Fig. 5, coupled with H in the y direction, E in the z direction, and solutions of the form $e^{j(\omega t - kx)}$ leads from Eq. (9) to the condition

$$j^2 k^2 = \mu \sigma j \omega \quad (10)$$

which gives for the wave number k

$$k = \pm (\mu \sigma \omega)^{1/2} (1 - j)/\sqrt{2} \quad (11)$$

The group velocity $d\omega/dk$ is proportional to $\sqrt{\omega}$, so the wave is highly dispersive, and the magnitude of the group velocity is very low, being only a few meters per second, even at 20 kHz. Although Eq. (11) allows two signs for k , only the attenuating solution is permissible, and recalling the definition of the skin depth δ in Eq. (1), we have, using the positive sign solution,

$$\begin{aligned} H_y(x, t) &= H_0 e^{j\omega t} e^{-j(1-j)x/\delta} \\ &= H_0 e^{j(\omega t - x/\delta)} e^{-x/\delta} \end{aligned} \quad (12)$$

where H_0 is the field amplitude at the surface of the medium. A similar equation describes the electric field E_z , and hence the current density in the z direction. Note that the "wave" attenuates ($e^{-x/\delta}$) at the same spatial rate as it oscillates ($e^{-jx/\delta}$). Outside the medium, for $x < 0$, the E and H fields have a more normal nonattenuating character, but the phase angle between E and H reflects the complex dissipative nature of the wave in the conducting medium. We note that the current density J_z also falls off as $e^{-x/\delta}$, essentially confining the current to a skin depth.

Our model has a voice-coil sheet of infinite length placed adjacent to the conducting magnetic medium, and we can characterize it with n turns per unit length. To compute the impedance per unit length of the coil (y direction), and per unit length of the surface along the wire (z direction), we first calculate η , the total induced flux per unit length of the wire,

$$\begin{aligned} \eta &= \int_0^l \int_0^\infty \mu H(x, 0) dx dz \\ &= \mu \int_0^\infty H_0 e^{-(1+j)x/\delta} dx \\ &= \mu H_0 \frac{\delta}{1 + j} \end{aligned} \quad (13)$$

The surface field H_0 is related to the current I in each turn of the wire by

$$H_0 = nI$$

Hence the inductance L per unit length along the coil per unit length along the wire is

$$\begin{aligned} L &= \frac{n\eta}{I} \\ &= n^2 \frac{\mu \delta}{1 + j} \end{aligned} \quad (15)$$

After multiplication by $j\omega$ to give impedance, and using Eq. (1) to express the frequency dependence, we have, for the impedance Z per unit length of coil per unit length along the wire,

$$Z = n^2 \sqrt{\frac{\mu j \omega}{\sigma}} \quad (16)$$

To obtain the actual impedance of a finite iron-filled coil of length L and radius r_0 , neglecting end corrections, we would multiply the result above by $L2\pi r_0$. Note that the impedance Z depends on $\sqrt{j\omega}$, which is the square root of a purely inductive behavior. As seen in earlier figures, the $\sqrt{\omega}$ dependence with attendant 45° phase shift is a common component of driver impedances. We shall call such behavior *semi-inductive*, and use the term *semi-inductor* to describe a component having an impedance given by

$$Z = K\sqrt{j\omega} \quad (17)$$

where K is the *semi-inductance* in *semihenrys*.

There is generally some small spacing between the voice coil and the magnetic pole structure, so that the flux in this region is precisely in phase with the coil current, as in normal inductive behavior. Thus we would expect a small but insignificant truly inductive component to exist in series with a dominant semi-inductive component. A more serious objection to the theory above concerns the range of the x integration in Eq. (13), since in real drivers the pole size is not infinite. As the frequency reduces, δ increases, and a local theory will no longer apply. In fact, Fig. 4 shows normal inductive behavior at the lowest frequencies. In an attempt to better model the actual impedance, we now turn to a cylindrical geometry which more closely resembles a real driver.

2 A CYLINDRICAL MODEL

Consider now a conducting cylindrical pole of radius r_0 , wrapped with a circumscribed voice coil, again resulting in an oscillatory tangential magnetic field along the cylinder axis. This problem is germane to understanding an induction furnace and has a well-known solution [8], which is sketched below.

Using a standard cylindrical coordinate system with z along the cylinder axis, r the distance from this axis, and ϕ the azimuthal angle, we ignore any ϕ and z variation in the magnetic field H . After an analysis, the resulting equation for $\mathcal{H}$, the field in the z direction,

is

$$\frac{\partial^2 \mathcal{H}}{\partial r^2} + \frac{1}{r} \frac{\partial \mathcal{H}}{\partial r} = \mu \sigma \frac{\partial \mathcal{H}}{\partial t} \quad (18)$$

which represents the counterpart of Eq. (9).

For solutions of $\mathcal{H}$ having a time dependence $e^{j\omega t}$, Eq. (18) becomes, for the time-independent field H ,

$$\frac{d^2 H}{dr^2} + \frac{1}{r} \frac{dH}{dr} - j\mu\sigma\omega H = 0 \quad (19)$$

which is a Bessel differential equation of zeroth order. A unique solution for Eq. (19), valid for values of r including $r = 0$, is

$$H = AJ_0(kr) \quad (20)$$

where $k^2 = -j\mu\sigma\omega$, equivalent to Eq. (10), and A is a constant. Note that if we write $K^2 = \mu\sigma\omega$, then $k^2 = -jK^2 = j^3 K^2$. Thus $k = j^{3/2}K$, where K is real. Bessel functions J_n with complex argument $j^{3/2}Kr$ are themselves complex, and are designated by the Kelvin functions ber_n and bei_n , having modulus M_n and phase ψ_n . Thus

$$\begin{aligned} J_0(kr) &= J_0(j^{3/2}Kr) \\ &= \text{ber}_0(Kr) + j \text{bei}_0(Kr) \\ &= M_0(Ke) e^{j\psi_0(Kr)} \end{aligned} \quad (21)$$

Since H_0 represents the magnetic field at $r = r_0$, we have

$$H(r) = H_0 \frac{J_0(j^{3/2}Kr)}{J_0(j^{3/2}Kr_0)} \quad (22)$$

The magnetic field has only a z component, and when Eq. (3), without the $\partial D/\partial t$ term, is expressed in cylindrical coordinates, we have for the current density J in the tangential direction

$$J = -\frac{\partial H}{\partial r} \quad (23)$$

Using Eq. (22), this gives

$$J(r, t) = j^{3/2}KH_0 \frac{J_1(j^{3/2}Kr)}{J_0(j^{3/2}Kr_0)} e^{j\omega t} \quad (24)$$

Physically, this current density should be maximum at the surface, while in addition for large Kr_0 (i.e., the radius quite a bit larger than a skin depth, which occurs at high frequencies) it should fall off as $e^{-(r_0-r)/\delta}$, just like H and J (or E) for the semi-infinite case, as exemplified by Eq. (12). In fact both H and J have this character, since for large Kr , both $J_0(j^{3/2}Kr)$ and

$J_1(j^{3/2}Kr)$ are asymptotic to $e^{Kr/\sqrt{2}}/\sqrt{2\pi Kr}$, ignoring phase factors. By considering the asymptotic values of the numerator and denominator in each of Eqs. (22) and (24), it is easily shown that both H and J contain factors of $e^{-(r_0-r)/\delta}$, and further analysis shows that the phase angles agree as well.

At very low frequencies, $Kr \ll 1$, $J_1(j^{3/2}Kr) \sim j^{3/2}Kr/2$, and $J_0(j^{3/2}Kr) \sim 1$, so that from Eq. (24) the current density J is zero at $r = 0$ and rises linearly toward the edge. The field H is constant over the cross section of the cylinder, and a simple analysis considering the flux in a circle of radius r , acting on the peripheral conducting material of length $2\pi r$, gives the above result also.

The magnetic flux Φ through the cylinder is obtained by integrating $\mu H(r)$ over the cross-sectional area of the cylinder,

$$\begin{aligned} \Phi &= \mu \int_0^{r_0} H(r) 2\pi r dr \\ &= \frac{2\pi\mu H_0}{J_0(j^{3/2}Kr_0)} \int_0^{r_0} r J_0(j^{3/2}Kr) dr \\ &= \frac{2\pi\mu H_0 r_0}{j^{1/2}K} \left[\frac{J_1(j^{3/2}Kr_0)}{j J_0(j^{3/2}Kr_0)} \right] \end{aligned} \quad (25)$$

The factor in square brackets in Eq. (25) asymptotically approaches unity as $Kr_0 \rightarrow \infty$, thus explaining the separation of the $j^{3/2}$ into $j^{1/2}$ and j factors. For $Kr_0 < 1$, this same factor varies as $j^{1/2}Kr_0/2$. Fig. 6 shows a plot of magnitude R and phase angle ϕ of the square brackets in Eq. (25), as a function of Kr_0 , derived from published tables [9]. Note that $K \propto \sqrt{\omega}$, so that the square of the horizontal coordinate is proportional to frequency. We relate H_0 to the current I and turns density n of the voice coil, as in Eq. (14), and the impedance Z per unit length of the coil becomes, after some manipulation,

$$Z = n^2 2\pi r_0 \sqrt{\frac{\mu j \omega}{\sigma}} R e^{j\phi} \quad (26)$$

At high frequencies, $Kr_0 \gg 1$ and $R e^{j\phi} \sim 1$, so that

$$Z \cong n^2 2\pi r_0 \sqrt{\frac{\mu j \omega}{\sigma}} \quad (27)$$

which represents semi-inductive behavior, just like Eq. (16), in which the circumference of the pole cylinder $2\pi r_0$ is explicitly included. When $Kr_0 \ll 1$, $R e^{j\phi} \sim j^{1/2}Kr_0/2$, and using $K^2 = \mu\sigma\omega$, Eq. (26) reduces to

$$Z \cong n^2 \mu \pi r_0^2 j \omega \quad (28)$$

which is normal inductive behavior, with value proportional to the pole cross-sectional area.

Fig. 6 shows that the transition between inductive and semi-inductive behavior is quite abrupt and occurs

when $Kr_0 = 2$, which translates to

$$\delta = \frac{r_0}{\sqrt{2}}. \quad (29)$$

However, Fig. 4, which already has the dc resistance removed from the measurement, shows that the transition is quite gradual. In addition, when realistic numbers for σ ($10^7 \Omega^{-1}/\text{m}$) and μ ($3000\mu_0$) are used for a normal iron pole of radius 10 mm, the condition Eq. (29) is obtained for a frequency of 0.17 Hz, whereas the real measurement shows a gradual transition at over 100 Hz. The author believes that these bothersome discrepancies can be resolved qualitatively by a consideration of the actual driver magnetic geometry.

Consider the typical driver structure shown in cross section in Fig. 7. Our model differs in a number of ways from an infinite iron cylinder wrapped with an infinite-length coil. The coil is short and usually rather squat, although the iron central pole is somewhat longer. But there is an air gap between the central pole and the top plate, which greatly reduces the net magnetic effect of the coil current. In addition the ferrite magnet has very low differential permeability, since it is a hard magnetic material, and acts like another large air gap. There may also be saturation effects in the iron near the voice-coil gap. The net result is that the effective permeability of the iron may be reduced by over a thousandfold, and this raises the effective transition to much higher frequencies. If the effective relative permeability is only 3, for example, but the conductivity

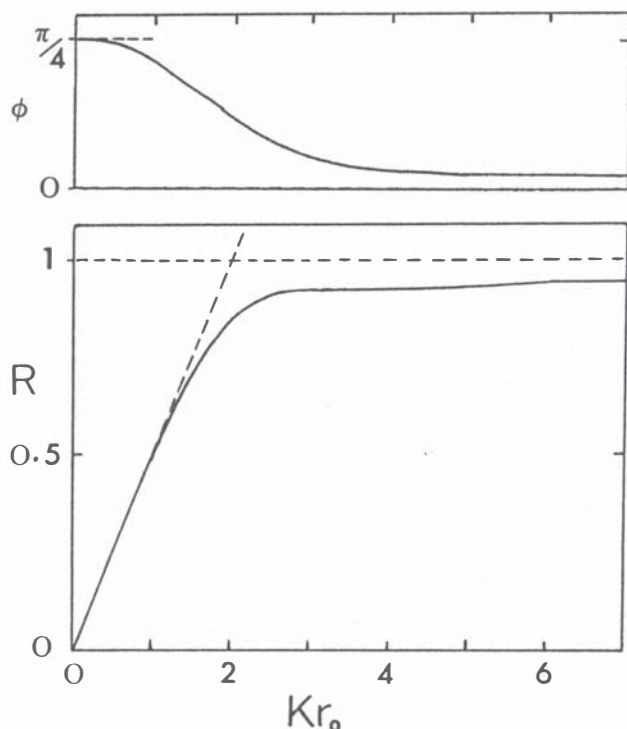


Fig. 6. Magnitude R and phase angle ϕ of Kelvin functions that occur in the theory of the skin effect for a cylinder, plotted versus Kr , where r is the radius from the center of the cylinder and K is defined in the text. The two dashed lines show the asymptotic behavior of magnitude R .

and the pole radius are the same as considered earlier, the transition occurs at about 170 Hz.

To lend credence to the above plausibility arguments, an iron toroid (actually a forged steel eye bolt) of 17.5-mm minor diameter and 65-mm major diameter was close wound with 24-gauge enameled magnet wire. Fig. 8 shows its measured impedance at low frequencies, with the dc coil resistance of 0.67Ω removed from the plot. Note that the semi-inductive behavior extends down to a few hertz, which is quite low for this steel sample of unknown permeability and conductivity, and supports the basic idea that when there are no demagnetizing air gaps, the inductive to semi-inductive transition is at a very low frequency.

3 SOME CONSEQUENCES OF THE EDDY CURRENTS

We have seen how the eddy currents in the pole structure of an electrodynamic driver cause basically a $\sqrt{j\omega}$ semi-inductive series impedance. Such frequency-dependent impedance variation causes a variation in the applied force when the driver is driven by a source of low output impedance. One thought might be that to reduce this effect, the eddy currents that circulate around the pole iron should be reduced. This could be accomplished by making the pole out of a nonconducting magnetic material, or by cutting slots diametrically across it to reduce the currents. Thus reducing the eddy currents allows the magnetic field to enter the pole, with a consequent increase in the inductive back EMF. The impedance would actually rise and make the driver less efficient. Eq. (27) shows that for decreased conductivity σ , the impedance would rise. Sufficient decrease of σ would revert the $Kr_0 > 1$ condition to $Kr_0 < 1$, and the impedance would behave with the normal inductive character of Eq. (28).

Thus it is somewhat ironic that the eddy currents which result in semi-inductive behavior are actually helpful; they reduce the underlying impedance from simple $j\omega$ dependence to $\sqrt{j\omega}$ dependence. How can we foster larger eddy currents to make the driver still better? Some manufacturers have used copper conducting rings around the central pole. Fig. 9 shows the blocked-cone impedance of a wide-range 8.5-in (216-mm) Philips driver that has been engineered to have an impedance dominated by the dc resistance of the coil. This driver is quite efficient and would have had an impoverished specification as a wide-range driver if the inductive rise had continued as shown in some of the earlier figures. It is not apparent at first glance that copper would be better than iron in the central pole. Although its conductivity may be up by an order of magnitude, its permeability is lower by a factor of over 1000, having nearly the same permeability as a vacuum. However, we have seen that the effective permeability of the iron is greatly reduced by the demagnetizing effects of the air gaps, and thus copper fares better than might be expected. The skin depth in copper is about 2 mm at 1 kHz, so for all higher audio

frequencies the currents would be confined quite close to the surface. On the other hand, the copper in the gap area reduces the static magnetic field, which reduces the efficiency of the driver.

Another aspect of the voice coil is the former on which it is wound. This is often an aluminum sheet with a slot to prevent circumferential eddy currents, as shown in Fig. 10(a). Heat from ohmic coil losses is spread by the former over a larger surface area for increased thermal performance. But the effect of the eddy currents is not totally removed by the slot. There are also eddy currents induced into a conducting former by its motion in the static field of the driver. Fig. 10(b) shows the coil former, opened out to be flat for easy pictorial representation. Suppose at a particular instant that it is moving upward as indicated by the arrow labeled v on the diagram. The crosshatched area represents the field B of the permanent magnet, shown drawn down into the plane of the figure. The former material experiences a downward rate of change of field dB/dt , as marked by $+$ signs, and an upward dB/dt as labeled by $-$ signs. This changing field causes the current flow lines as drawn on the figure, which

cause a net downward drag force on the aluminum former, as expected from the interaction of the currents in the area containing the static field B . What is less apparent is that this distributed force is not uniform along the circumferential width of the former due to the divergence of the induced current at each end of the sheet, as shown by the force diagram in the lower part of Fig. 10(b). Thus there is a tipping or a canting force induced on the voice coil that is proportional to the cone velocity. This small but deleterious effect can be avoided by using two slots in the former, placed diametrically opposite each other. This also reduces the net retarding damping force.

The aluminum coil former also has a slight secondary effect. Fig. 11 shows the impedance (with dc resistance removed) of a 1-in (25.4-mm) voice coil with aluminum former, measured by itself away from any part of the driver magnetic structure. There is some deviation from a straight-line inductive behavior at high frequencies, due to the eddy currents in the former, which become significant when the skin depth decreases and becomes comparable to the former thickness. These currents exist because there is a tendency for the field lines to

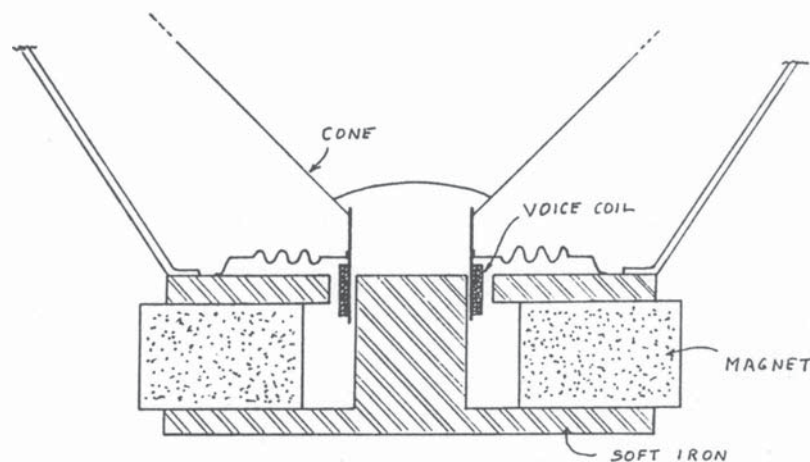


Fig. 7. Cross section of typical low-frequency drive unit, showing magnetic components and voice coil. The air gap for the coil and the low differential permeability of the ceramic magnet greatly reduce the effective permeability of the iron.

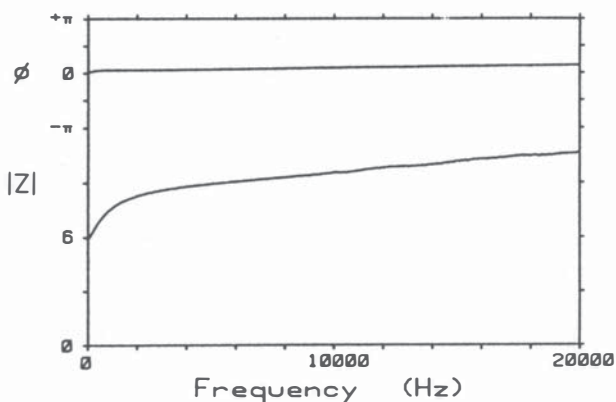


Fig. 8. Impedance magnitude and phase angle for iron toroid closely wrapped with insulated wire. The dc coil resistance has been subtracted out of the measurement and the impedance has $\sqrt{f}$ character to very low frequencies, testifying to the validity of the theory if there are no demagnetizing air gaps.

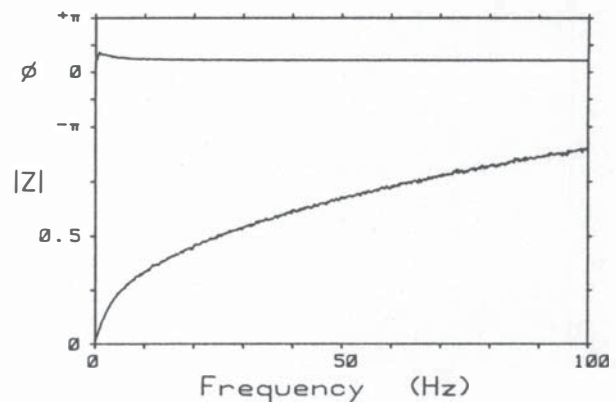


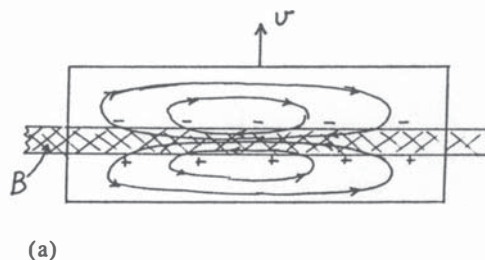
Fig. 9. Measured impedance magnitude and phase angle for 8.5-in (216-mm) wide-range loudspeaker with controlled impedance characteristic. Controlling the impedance rise by using conducting rings around the central pole increases the efficiency of the motor structure.

have a component normal to the former, especially since it is often considerably longer than the coil winding.

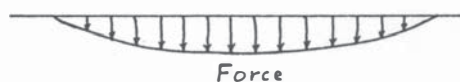
4 OTHER ASPECTS OF THE EDDY CURRENTS

It is common practice to measure driver parameters by measuring their resonance characteristics both in free air and when mounted on a test box. To what extent does the blocked-cone impedance affect the results of such measurements? To answer this question, a computer simulation was carried out with two limiting conditions for the blocked-cone impedance, (1) resistive and (2) purely semi-inductive. The cone velocity generates a back EMF which gives an image of the acoustomechanical fundamental resonance in the measured impedance. A simple analysis of a damped harmonic oscillator shows that the resonance in the impedance can be characterized by the frequency of resonance ω_0 , the Q factor, and the peak of the impedance $Z_{\max}$. Our model for Z is then

$$Z(\omega) = Z_{bl}(\omega) + Z_{\max} \frac{j\omega\omega_0/Q}{\omega_0^2 - \omega^2 + j\omega\omega_0/Q} \quad (30)$$



(a)



(b)

Fig. 10. (a) Shape of typical aluminum voice-coil former. (b) Flattened pictorial representation of coil former in (a), with crosshatched area representing field in coil gap. As the former moves upward, eddy currents flow as illustrated, producing a nonuniform distributed force which tends to tip the voice coil.

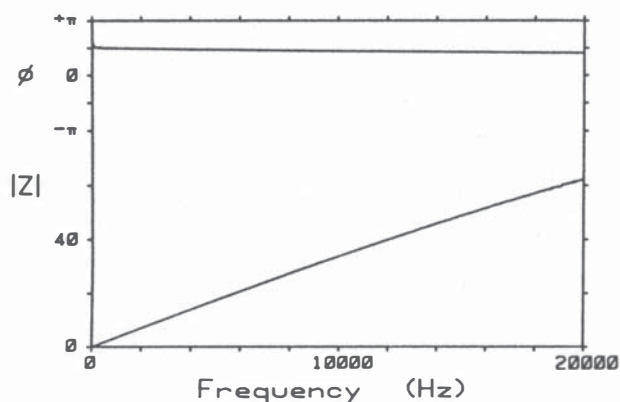


Fig. 11. Impedance magnitude and phase angle for 1-in (25.4-mm)-diameter voice coil with aluminum former, measured by itself away from any magnetic material. The dc coil resistance of 4.6 Ω has been subtracted out to display clearly the effects of the coil former at higher frequencies.

where $Z_{bl}(\omega)$ is either a constant resistance R or a purely semi-inductive impedance as given by Eq. (17). Fig. 12 shows curves for the two cases. In each case the added peak impedance was 20 Ω , the Q was 4, and the background constant or semi-inductive impedance had a magnitude of 8 Ω at the resonance frequency of 100 Hz. It is clear that the semi-inductive background phase of 45° is responsible for the slightly lower net total peak and the shortfall of the impedance on the high side of the resonance. This example shows that a proper measurement of driver parameters requires that the impedance be reasonably well modeled near the resonance.

A very significant aspect of loudspeaker drivers is their thermal design. Although the voice-coil absorbed power is simply given by $R_{dc}I_{rms}^2$, the total power absorbed by the driver can well exceed this due to the eddy currents in the central pole, which form a kind of crude secondary winding of a transformer in which the currents are confined to a skin depth. The current sheet is confined more and more as frequency rises, causing ever-increasing dissipation in the iron. The driver whose impedance is shown in Fig. 1, for example, has a dc voice-coil resistance of 5.7 Ω , but at 5 kHz

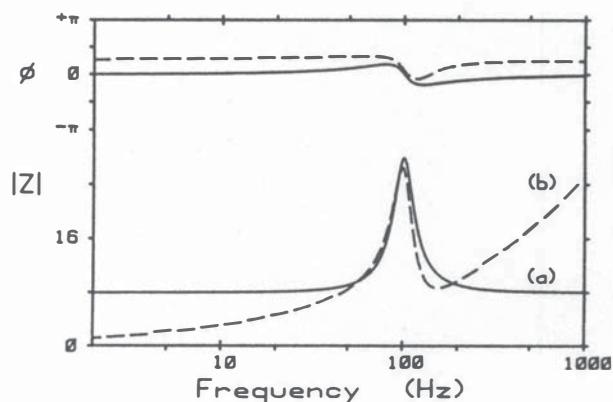


Fig. 12. Computer simulation of driver impedance curve with two limiting conditions for blocked-cone impedance. (a) Constant 8 Ω resistive. (b) $\sqrt{jf}$ behavior with 8- Ω magnitude at resonance frequency of 100 Hz. The electrical Q of the driver is 4 and the resonant peak has a height of 20 Ω .

its impedance is $45\ \Omega$, with a phase angle of nearly 45° . If excited by a sine wave at 5 kHz, an rms current of 1 A would cause a thermal dissipation of 5.7 W in the voice coil, while the total power input is $45 \cos 45^\circ = 31.8$ W. Thus 26.1 W is dumped directly into the central pole surface (within a skin depth), and this severely modifies the long-term power-handling capability of a driver due to heating of the whole magnet assembly. It is not clear whether designers in general have realized that due to the transformer action, the magnet structure is separately heated without this heat having to flow through the voice-coil thermal resistance. When power transients are applied and resulting coil temperatures are measured in an attempt to measure the coil thermal inertia and the thermal resistance or time constant, the *actual* physical values will not correspond to the naively measured values if consideration is not given to the eddy currents. It is even necessary to consider the signal spectrum in such tests, since the fraction of the power transformed into the pole structure is strongly dependent on frequency.

How would we reduce the energy absorbed directly by the pole structure? A conductive ring may help, although it compromises other characteristics as outlined earlier. If room-temperature superconducting materials become a reality, a thin layer of such material around the pole would greatly reduce the semi-inductive absorbed power, lowering the impedance over a wide frequency range. In fact a superconducting voice coil alone would represent only a partial solution, since the bulk of the power is transformed into the pole directly at higher frequencies.

Compensation of the loudspeaker electric impedance, making it solely resistive, helps amplifier driving ability and maintains system flatness even in the face of line resistance. How would a semi-inductor be compensated? Consider a system made of a series resistor R (voice coil) and a semi-inductive impedance $K\sqrt{j\omega}$, as given by Eq. (17). What parallel impedance would make the combination appear as a resistor R ? It is easily shown that the parallel impedance $Z_{||}$ must have a form

$$Z_{||} = R + \frac{R^2}{K\sqrt{j\omega}} \quad (31)$$

which represents a resistor R in series with what we could term a *semicapacitor*. This latter component has a -45° phase angle and an $\omega^{-1/2}$ magnitude dependence of impedance. A transmission line modeled with a distributed shunt capacitance and series resistance, called a diffusion line or Kelvin cable, has an input impedance that is precisely a semicapacitor. In practice the compensation might proceed with multiple parallel RC branches, much like a portion of a pink-noise equalizing filter. If a semi-inductor is paralleled with a semicapacitor, the magnitude of the impedance has a broad maximum, not a sharp one, hence the combination only *semiresonates*. Observations pertinent to compensating networks can be found in [10].

Another question of interest is the nature of the current

waveform when a step or an impulse voltage is applied to a semi-inductor. Physically, for an applied voltage step, the coil will try to create a magnetic field which takes a while to diffuse into the iron. Hence there will be no back EMF for the first instant, and the current waveform will rise sharply at the leading edge. A Laplace transform analysis of the voltage-current relationship for a semi-inductor (for which $j\omega$ in the impedance is replaced by the Laplace variable s) gives

$$V(s) = K \sqrt{s} I(s) \quad (32)$$

For a step voltage input of amplitude V_0 applied at $t = 0$, $V(s) = V_0/s$, and thus

$$I(s) = \frac{V_0 K}{s^{3/2}} \quad (33)$$

for which the inverse Laplace transform gives the current waveform

$$i(t) = \frac{2V_0 K}{\sqrt{\pi}} \sqrt{t} \quad (34)$$

It is not surprising to see $\sqrt{t}$ relationships in situations where diffusion is involved. If an impulse of area V_0 is applied to a semi-inductor at $t = 0$, the current waveform takes the form

$$i(t) = \frac{V_0 K}{\sqrt{\pi t}} \quad (35)$$

which represents the time derivative of the right-hand side of Eq. (34).

If the semi-inductor has some series resistance, then none of the simple analytic relationships apply. Fig. 13 shows a measurement of the current response to an impulse for the iron toroid whose impedance is shown in Fig. 8. The peak of the curve has been clipped to better show the shape of the decaying portion, which is similar to Eq. (35).

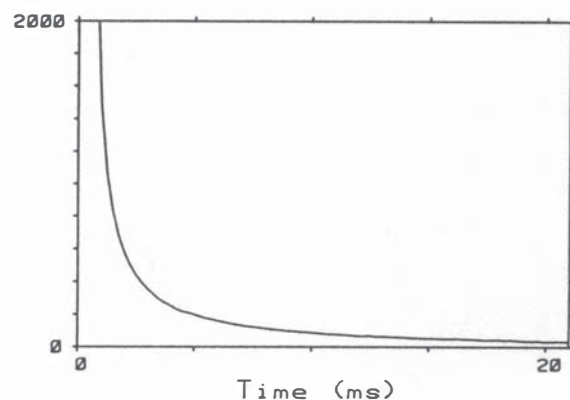


Fig. 13. Current waveform for iron toroid whose impedance is shown in Fig. 8. The excitation is a voltage impulse and the peak of the curve has been clipped to better show the shape of the decaying portion of the waveform, which is similar to $t^{-1/2}$.

5 SUMMARY

We have seen how eddy currents in the pole structure are responsible for the $\sqrt{j\omega}$ impedance behavior of an electrodynamic driver. The theories presented show overall agreement with measurement and lead to an understanding of related features in the driver.

6 ACKNOWLEDGMENT

This work was supported by the Natural Sciences and Engineering Research Council of Canada. The author is grateful to Richard Small for pointing out the work of one of his students [4]. Ewan MacPherson has helped in plotting the impedance data.

7 REFERENCES

- [1] L. L. Beranek, *Acoustics* (McGraw-Hill, London, 1954); reprinted by Amer. Inst. of Physics, New York, 1986, p. 185 ff.
- [2] A. N. Thiele, "Loudspeakers in Vented Boxes: Part 1," *Proc. IREE (Australia)*, vol. 22, pp. 487–508 (1961 August); reprinted in *J. Audio Eng. Soc.*, vol. 19, pp. 382–392 (1971 May); see Fig. 4.
- [3] M. J. Hawksford, "The Essex Echo. Unification: Track One," *HiFi News Rec. Rev.*, p. 53 ff. (1986 May).
- [4] I. M. Dash, "An Equivalent Circuit Model for the Moving Coil Loudspeaker," honours thesis 1982, School of Electrical Engineering, University of Sydney, Australia.
- [5] D. D. Rife and J. Vanderkooy, "Transfer Function Measurement Using Maximum-Length Sequences," presented at the 83rd Convention of the Audio Engineering Society, *J. Audio Eng. Soc. (Abstracts)*, vol. 35, p. 1064 (1987 Dec.), preprint 2502.
- [6] J. Borish and J. B. Angell, "An Efficient Algorithm for Measuring the Impulse Response Using Pseudorandom Noise," *J. Audio Eng. Soc.*, vol. 31, pp. 478–488 (1983 July/Aug.).
- [7] J. Vanderkooy, "Contactless Hall Coefficient Measurement in Materials of Small Hall Angle in the Skin Depth Regime," *Rev. Sci. Instrum.*, vol. 49, pp. 1166–1168 (1978 Aug.).
- [8] J. Lammeraner and M. Staffl, *Eddy Currents* (Iliffe, London, 1966), p. 88 ff.
- [9] M. Abramowitz and I. A. Stegun, *Handbook of Mathematical Functions* (Dover, New York, 1968), p. 432.
- [10] A. N. Thiele, "Optimum Passive Loudspeaker Dividing Networks," *Proc. IREE (Australia)*, vol. 36, pp. 220–224 (1975 July).

THE AUTHOR



John Vanderkooy was born in The Netherlands in 1941 and received all of his education in Canada, earning a B.Eng. degree in engineering physics in 1963 and a Ph.D. in physics in 1967, both from McMaster University in Hamilton, Ontario. For some years he followed his doctoral interests in low-temperature physics of metals at the University of Waterloo, where he is currently an associate professor of physics. His research is now mainly in electroacoustics.

A fellow of the AES and a member of the IEEE, Dr.

Vanderkooy has contributed a series of papers at conventions and to the *Journal*, many in collaboration with his colleague Stanley Lipshitz. Together with a number of graduate students they form the Audio Research Group at the University of Waterloo.

Dr. Vanderkooy's current interests are digital audio signal processing, measurement of transfer functions with maximum-length sequences, diffraction of loudspeaker cabinet edges, and the role of eddy currents in loudspeaker drivers.